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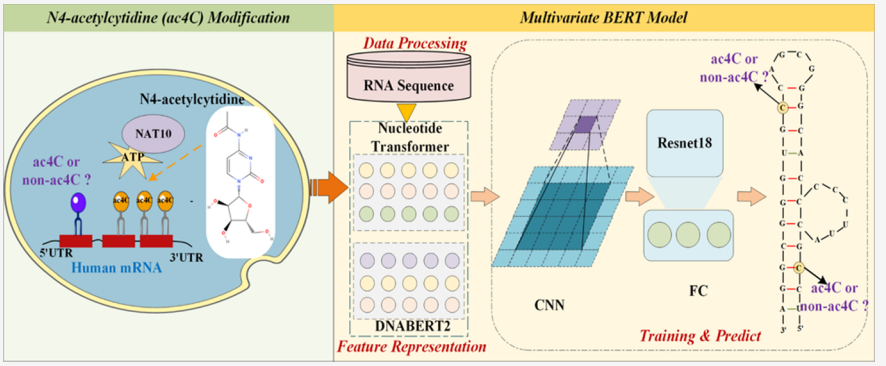
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NBCR-ac4C: A Deep Learning Framework Based on Multivariate BERT for Human mRNA N4-Acetylcytidine Sites Prediction

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Cite This: J. Chem. Inf. Model. 2024, 64, 8074−8081 Read Online

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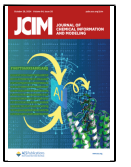
ABSTRACT: N4-acetylcytidine (ac4C) plays a crucial role in regulating cellular biological processes, particularly in gene expression regulation and disease development. However, experiments to identify ac4C in a wet lab are time-consuming and costly, and the learning-based methods struggle to capture the underlying semantic knowledge and relations within sequences. To address this, we propose a deep learning approach called NBCR-ac4C based on pretrained models. Specifically, we employ Nucleotide Transformer and DNABERT2 to construct contextual embedding of nucleotide sequences, which effectively mine and express context relations between different features in the sequence. Convolutional neural network (CNN) and ResNet18 are then applied to further extract shallow and deep knowledge from context embedding. Depending on extensive experiments for the prediction of ac4C sites in nucleotide sequences, we observe that NBCR-ac4C outperforms general learning-based models. It achieves the highest accuracy (ACC) of 83.51% and an Area Under the Receiver Operating Characteristic Curve (AUROC) of 89.58% on an independent test set. Moreover, the proposed model, compared to the current state-of-the-art (SOTA) model LSA-ac4C, demonstrates higher ACC and AUROC by 0.81−3.7% and 0.05−1.58%, respectively. The data set and code are available on https://github.com/2103374200/NBCR to facilitate further discussion on NBCR-ac4C.

1. INTRODUCTION

The MODOMICS database records more than 170 types of ribonucleic acid (RNA) modifications catalyzed by specific enzymes.[1−3] N4-acetylcytidine (ac4C) involves attaching an acetyl group to the N4 position of cytidine, thus, setting it apart from other RNA modifications. Studies indicate that ac4C significantly impacts RNA stability and interactions.[4] It not only participates in biological processes like RNA-protein interactions but also shows close associations with various disease treatments. For instance, ac4C has been demonstrated to facilitate the treatment of neuropathic pain,[5] prevent bone loss caused by oophorectomy,[6] and exhibit significant efficacy in cervical cancer therapy.[7] Therefore, precise identification of ac4C holds crucial significance for research in the biology, medicine, and molecular biology fields.

Currently, detection and identification of ac4C include experimental methods, as well as bioinformatics methods based on machine learning or deep learning. A typical experimental method is the ac4Cseq technique,[8] which identifies ac4C through a chemical reaction between ac4C and sodium cyanoborohydride. However, this experimental method (wet lab) suffers from drawbacks, such as time consumption, resource intensiveness, and expensive materials. Hence, machine learning or deep learning methods independent of wet lab (with high cost) have been applied for ac4C prediction. The latest machine learning models, including iRNA-ac4C,[9] stacking-ac4C,[10] and ac4C-AFL,[11] demonstrate distinct advantages in ac4C site identification. iRNA-ac4C employs

Received: August 6, 2024
Revised: September 6, 2024
Accepted: September 25, 2024
Published: October 5, 2024



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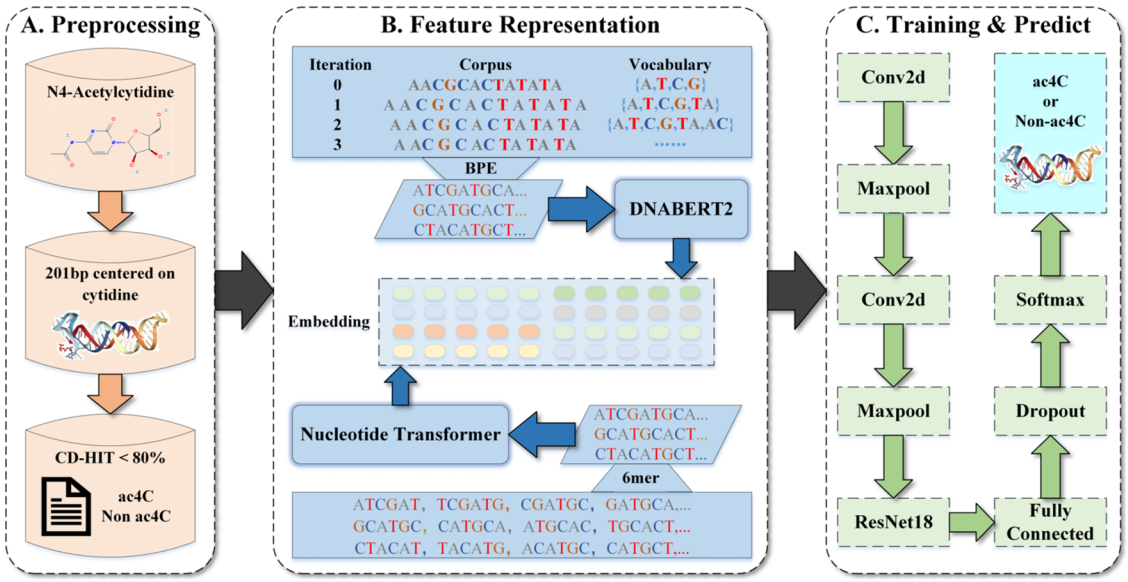


Figure 1. Overview of the NBCR-ac4C.

Gradient Boosting Decision Trees (GBDT) for ac4C recognition, achieving a high accuracy. Stacking-ac4C combines three encoding methods (Kmer, PseKNC, and PseEIIP) and utilizes a stacking-based algorithm for classification, leading to more precise ac4C site identification. ac4C-AFL identifies the optimal feature set from 16 sequence-derived descriptors and uses these features to build 176 baseline models. Through a two-step feature selection process, the most effective model for ac4C recognition was determined. The deep learning model LSA-ac4C[12] integrates dual-layer long short-term memory (LSTM) and self-attention mechanisms to achieve high-precision prediction of ac4C sites.[13,14] Li et al.'s MetaAc4C[15] model utilizes pretrained bidirectional encoder representations from a Transformer. The model is based on a Bidirectional Long Short-Term Memory (BLSTM) architecture, combining attention mechanisms and residual connections to predict ac4C sites. Nevertheless, these learning-based models have shortcomings in understanding the inherent sequential and functional complexity of nucleotide sequences, which may affect the accurate localization of ac4C sites.

This study leverages BERT-based pretrained models to learn semantic correlations of sequences and designs residual network to extract diversified semantic knowledge within sequences. Based on this, the NBCR-ac4C deep learning network model is developed to improve the analysis and prediction of ac4C. It abandons traditional manual feature design and instead employs two pretrained BERT models: Nucleotide Transformer and DNABERT2 were used for feature extraction. This approach can acquire contextual embedding of multivariate BERT to better capture sequence semantic relations.[16] The reason for choosing these pretrained models is the extensive application of BERT in bioinformatics research. For example, by learning and understanding various relations in large-scale nucleotide sequences, BERT models can not only improve the accuracy of promoter recognition[17] but also enhance the detection results of DNA enhancers.[18]

The ac4C sequence feature vectors preprocessed by the constructed pretrained models are fed into a convolutional neural network (CNN), to extract shallow features, achieve feature scaling, and ensure alignment of the feature space.[19] Subsequently, NBCR-ac4C deploys ResNet18 to gain deep semantic knowledge of nucleotide sequences, including multiple patterns and diverse dependencies. This approach aims to reveal correlations between specific sequences or sites and ac4C modification functions, thereby improving the identification of ac4C sites.[20] In summary, the proposed NBCR-ac4C successfully merges feature representation (based on BERT pretrained models), extraction of shallow semantic knowledge (operations of convolution and pooling), and extraction of deep semantic knowledge (correlations obtained by ResNet18). This not only effectively improves the performance of ac4C detection but also provides a basis for understanding the diverse relations between biological molecules.

To validate the presentation and identification capabilities of NBCR-ac4C for nucleotide sequences, we conducted a large number of comparative experiments, containing various deep learning models, including the SOTA model LSA-ac4C. Compared to deep learning models composed of the convolutional neural network or the recurrent neural network, NBCR-ac4C exhibits excellent performance on independent test sets with an accuracy (ACC) of 83.51% and an area under the receiver operating characteristic curve (AUROC) of 89.58%. This indicates its outstanding ability to distinguish between positive (ac4C) and negative (non-ac4C) samples even at different decision boundaries. Meanwhile, the Matthews correlation coefficient (MCC) of the NBCR-ac4C model (67.06%) surpasses that of the current SOTA model LSA-ac4C (65.66%), which further verifies the model's advantages in predicting results. In conclusion, compared with baselines, NBCR-ac4C demonstrates superior performance in representing complex nucleotide sequences, resulting in higher performance in detecting ac4C sites.

2. MATERIALS AND METHODOLOGY

The NBCR-ac4C model designed in this study is illustrated in Figure 1. It comprises three steps: data set preprocessing, feature representation, as well as model training and prediction. In Figure 1A, the preprocessing of nucleotide sequences involves using CD-HIT to remove sequence

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redundancy with a threshold set at 0.8. Figure 1B depicts the feature representation of nucleotide sequences, which employs pretrained models Nucleotide Transformer and DNABERT2 to construct multivariate BERT context embedding. In Figure 1C, the deep neural network consists of CNN and ResNet18 for extracting semantic information, following a fully connected layer for labeling ac4C site prediction results. Details of each step are presented below.

2.1. Data Set Preprocessing. This study utilized the well-known benchmark data set introduced by Su et al.,[9] which was originated from biological experiments and made up of 4250 candidate ac4C peaks. To ensure the reliability of this data set, only cytosines closest to the ac4C peaks were selected as modification sites. 100 nucleotides were extracted from both sides of these modification sites as positive samples. For the negative samples, cytosines were located at the center of the randomly chosen regions outside the peaks, which span 201 nucleotides at the same time. Meanwhile, the CD-HIT tool with a threshold of 0.8 was adopted to eliminate redundancy. We randomly selected an equal number of positive and negative samples in the experiments. The final data set was divided into a training set and an independent test set at a ratio of 4:1. The training set contains 2206 positive and 2206 negative samples, while the independent test set consists of 552 positive and 552 negative samples. The preprocessing of the data set is illustrated in Figure 1A.

2.2. Multivariate BERT Contextual Word Embeddings Representation. The long-range dependencies between different parts of the aforementioned nucleotide sequence (RNA data) are difficult to capture by using traditional feature extraction methods such as Kmer.[21] To address this, this study selects two pretrained models based on BERT (Bidirectional Encoder Representations from Transformers[22]), Nucleotide Transformer, and DNABERT2, to represent nucleotide sequence features.[23,24] Both are deep bidirectional neural networks based on the Transformer architecture, which handles the context knowledge in nucleotide sequences better through their self-attention mechanism. Additionally, these BERT-based models automatically learn the optimal feature representation from the nucleotide sequence. It further reduces the negative impacts (such as time costs) of manually selecting or designing features.

As shown in Figure1B, the nucleotide sequences are first sent into Nucleotide Transformer and DNABERT2 for processing, respectively. Specifically, the Nucleotide Transformer transforms a nucleotide sequence (length of 201) to a two-dimensional feature embedding with a shape of 198*4107, while DNABERT2 converts the nucleotide sequence into a different feature embedding with a shape of 198 × 768. The converted feature embedding is concatenated with a new shape of 198 × 4875. The purpose of this process is to leverage two different BERT models to get diverse relations of features from nucleotide sequences.

2.2.1. Nucleotide Transformer. The Nucleotide Transformer is a model trained on whole-genome nucleotide sequences. This model incorporates data from over 3200 human genomes as well as 850 genomes from various other species. In this study, the second version of the Nucleotide Transformer (e.g., nucleotide-transformer-v2-500m-multispecies) is applied to annotate the nucleotide sequence ac4C. It employs Rotary embeddings instead of general positional embedding,[25] and each attention layer and gated linear unit uses the unbiased swish activation function. Furthermore, since the Nucleotide Transformer can accommodate sequences of up to 2048 tokens, it generates longer context windows (12 kbp), resulting in enhanced generalization capabilities across multiple species data sets. The Nucleotide Transformer utilizes 6-mers as tokens to annotate the ac4C sequence studied in this paper (Figure 1B). The vocabulary size maintained by the tokenizer in this pretrained model is set to 4105, with a maximum token sequence length allowed of 1000. Upon inputting a nucleotide sequence of 198 bases into the Nucleotide Transformer, the output is a two-dimensional matrix with dimensions of 198 × 4107.

2.2.2. DNABERT2. DNABERT2, a Transformer-based pretrained model, is also an efficient way for genomic processing. Due to its utilization of the Attention Mechanism with Linear Bias (ALiBi), as well as strategies like FlashAttention and low-precision layer normalization, it effectively overcomes input length limitations and reduces the resources required for training (such as time and memory). In this work, the ac4C sequence is carried out through the Byte Pair Encoding (BPE) algorithm in DNABERT2, subsequently training and generating a variable-length token vocabulary for the entire genome data set, instead of taking use of the general k-mers method. As illustrated in Figure 1B, DNABERT2 learns the aforementioned token vocabulary based on the character frequencies in the corpus consisting of nucleotide sequences. Meanwhile, the tokenization capability of DNABERT2 enables this model to focus on understanding the inherent semantics of the sequences, which avoids feature space degradation because of different representations of the same input and enhances the expressive ability of features for each sample. Upon inputting a nucleotide sequence of 201 bases into DNABERT2, the output is a two-dimensional matrix with dimensions of 198 *768.

2.3. Model Training and Prediction. *2.3.1. Shallow-Level Feature Extraction.* The ac4C sequence features, preprocessed and concatenated through Nucleotide Transformer and DNABERT2, have a dimensionality of 198 × 4875. In this step, two convolutional layers (CNN) and two max-pooling layers (Maxpool) are employed to extract shallow-level knowledge from the sequence features. The CNN layers initially deal with the 198 × 4875 sequence features, leveraging sparse interactions and capabilities to capture translational invariance.[26] Meanwhile, the Maxpool layers aim to reduce the spatial dimensions of the input sequence features, which reduces model parameter size and enhances model generalization.[27]

In Figure 1C, the NBCR-ac4C model consists of two CNN layers and two Maxpool layers. The first CNN layer contains 1 input channel and 3 output channels, with a kernel size of (3, 2048) and a stride of (1, 1). The first Maxpool layer has a pooling window size of (1, 2) and a stride size of (1, 2). These layers transform the input sequence features (198 × 4875) into three different feature maps, each representing distinct information. The second CNN layer comprises 3 input channels and 3 output channels, with a kernel size of (3, 1026) and a stride of (1, 1). The parameters of the second Maxpool layer are identical with those of the first Maxpool layer. These layers further extract and combine features from the ac4C sequence, resulting in shallow-level features. After two layers of convolution and pooling, the 198 × 4875 two-dimensional matrix is transformed into a 194 × 194 two-dimensional matrix. They retain crucial sequence knowledge while compressing feature dimensions, leading to a decrease of the complexity of sequence processing.

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2.3.2. Deep-Level Feature Extraction. The sequence features of the two-dimensional matrix 194 × 194 will be passed into ResNet18 to perform deep-level feature extraction. As one of the classic residual networks, ResNet18 is devoted to addressing issues such as gradient vanishing in deep neural networks. It, therefore, further explores deep-level knowledge within the biological sequence.[20] The sequence features obtained from Section 2.3.1 will undergo a series of stacked residual blocks in ResNet18, composed of multiple convolutional layers, batch normalization layers, and skip connections. These are eventually transformed into a one-dimensional vector containing 1000 elements. It is precisely due to the special operations in ResNet18, such as skip connections to ensure the propagation of gradient flow information, that the one-dimensional vector accurately encodes the position and features of the original input data set in the feature space. We thus capture deep-level knowledge within the ac4C sequence.

2.3.3. Prediction and Training Strategy. The feature vector (1000-dimensional) generated by ResNet18 is first fed into a fully connected layer composed of 1000 neurons. Subsequently, regularization is applied via a Dropout layer with a size of 0.5 to prevent overfitting of the neural network.[28] When predicting the labels of the initial sequences ("ac4C" and "Non-ac4C"), we adopted the Softmax activation function to compute the probabilities for each sequence. The probability for "ac4C″ or "Non-ac4C" of each sequence ranges between 0 and 1, and the sum of them equals 1. The result with the highest value is considered the final prediction for the given input sequence.

During the training of NBCR-ac4C, the training set partitioned in Section 2.1 (2206 positive and 2206 negative samples) needs to be randomly split in a 9:1 ratio, with one portion reserved as the validation set to evaluate model performance, while the rest is used for model training. Additionally, the batch size is set to 4, and Cross Entropy Loss is regarded as the loss function, as follows

$$L(\theta) = \frac{1}{N}\sum_{i=1}^{N} y_i \log\left(h_\theta(x)\right) - (1 - y_i)\log\left(1 - h_\theta(x)\right) \tag{1}$$

where y_i is the label of the i-th sample, $h_\theta(x)$ is the output of the neural network, and N is the number of samples. The initial learning rate of the Adam optimizer is defined as 2e-6,[29] with the learning rate decay method chosen as LambdaLR. After 4 epochs of training, the learning rate decay of the Adam optimizer is adjusted to 2e-7. The converged NBCR-ac4C model is independently tested by the test set from Section 2.1, which consists of 552 positive and 552 negative samples.

All model training and testing in this work were conducted on an Ubuntu server equipped with an Intel(R) Xeon(R) Platinum 8336C processor and an NVIDIA GeForce RTX 4090 GPU.

2.4. Evaluation Metrics. To assess the model's performance, the following metrics selected for evaluation are Sensitivity (SEN), Specificity (SPE), Accuracy (ACC), Matthews Correlation Coefficient (MCC), and Area Under the Receiver Operating Characteristic curve (AUROC). SEN measures the model's ability to detect positives, while SPE evaluates its ability to exclude negatives. ACC represents the proportion of correctly predicted samples. MCC considers True Positive, True Negative, False Positive, and False Negative for a balanced evaluation. AUROC analyzes binary classification performance by true positive and false positive rates. Formulas 2−8 calculate these metrics. TP (True Positive) represents correct positive identifications; TN (True Negative) denotes correct negative identifications; FP (False Positive) refers to incorrect negative predictions as positives; and FN (False Negative) signifies incorrect positive predictions as negatives. TPR or Sensitivity quantifies the proportion of actual positives correctly predicted, while FPR calculates the proportion of actual negatives incorrectly predicted.

$$\mathrm{SEN} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{2}$$

$$\mathrm{SPE} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{3}$$

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{4}$$

$$\mathrm{MCC} = \frac{\mathrm{TP} \times \mathrm{TN} - \mathrm{FP} \times \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}} \tag{5}$$

$$\mathrm{AUROC} = \int \mathrm{TPR}\, d(\mathrm{FPR}) \tag{6}$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{7}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}} \tag{8}$$

3. EXPERIMENTS AND DISCUSSIONS

3.1. Rationality Validation of the Designed Deep Neural Network. In the training and prediction phases of NBCR-ac4C, a neural network is designed with convolutional and residual layers. This network handles feature representations from the Nucleotide Transformer and DNABERT2 (198 × 4875) context embedding, capturing both shallow and deep nucleotide sequence knowledge. To validate this architecture, models with Long Short-Term Memory (LSTM) or multihead self-attention (Att) are developed to predict ac4C sites. Note that LSTM manages long sequence dependencies,[30] while Att captures richer semantic information.[31]

Three deep learning models, LSTM + Att, CNN + Att, and CNN + LSTM + Att, are compared with NBCR-ac4C. As shown in Table 1, NBCR-ac4C achieves a SEN of 91.40% (5.88% higher than CNN + Att) and an ACC of 83.71%, with an MCC of 68.23% (2.08% higher than CNN + Att). These results indicate that NBCR-ac4C outperforms metrics like TP, TN, FP, and FN. Independent test set results (Table 2 and

Table 1. Prediction Results on Validation Set, Compared with Three Deep Learning Models[a]

model	SEN (%)	SPE (%)	ACC (%)	MCC (%)
LSTM + Att	83.71	79.64	81.67	63.40
CNN + Att	85.52	**80.54**	83.03	66.15
CNN + LSTM + Att	82.81	78.28	80.54	61.15
NBCR-ac4C	**91.40**	76.02	**83.71**	**68.23**

[a]The highest score in each column is shown in bold.

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Table 2. Prediction Results on Test Set, Compared with Three Deep Learning Models[a]

model	SEN (%)	SPE (%)	ACC (%)	MCC (%)
LSTM + Att	82.79	80.80	81.79	63.60
CNN + Att	82.25	80.98	81.61	63.23
CNN + LSTM + Att	**86.23**	78.26	82.25	64.70
NBCR-ac4C	84.96	**82.07**	**83.51**	**67.06**

[a]The highest score in each column is shown in bold.

Figure 2) further confirm NBCR-ac4C's superiority, achieving 83.51% ACC, 67.06% MCC, and 89.58% AUROC, compared to 81−82% ACC, 62−66% MCC, and 87−88% AUROC for the other models.

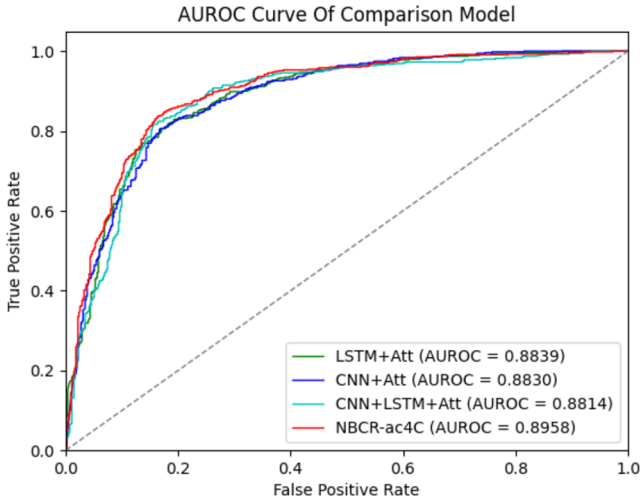


Figure 2. AUROC curves on the test set, compared with three deep learning models.

These results demonstrate that while LSTM and Att are effective for long sequences and semantic information, they are less effective for ac4C context embedding from the Nucleotide Transformer and DNABERT2. Instead, the CNN and ResNet18-based network excel by leveraging translational invariance (CNN) and ensuring gradient flow (ResNet18) to extract precise shallow and deep semantic information, enhancing ac4C identification accuracy.

3.2. Ablation Study. To further validate the impact of different components within NBCR-ac4C on ac4C recognition, ablation experiments are conducted. By removing modules from the model (Nucleotide Transformer, DNABERT2, CNN, and ResNet18), four comparative models are created: No-NT, No-DB2, No-CNN, and No-ResNet18.

Table 3 shows the ablation results on the validation set. Removing the Nucleotide Transformer significantly affects detection, reducing ACC from 68.23 to 58.66% and MCC from 65.62 to 55.02%. This impact is greater than removing DNABERT2, due to the Nucleotide Transformer's wider vocabulary and larger model size. Models without the second CNN layer or ResNet18 also perform worse than No-NT, highlighting the importance of feature representation, especially from the Nucleotide Transformer. Furthermore, the effect on the extraction of shallow (from CNN) and deep (from ResNet18) knowledge by the designed deep learning model is similar, with the MCC and ACC decreasing by approximately 3.31−3.9% and 1.36−1.58%, respectively. Nevertheless, it is hard to be replaced by LSTM, Att, etc., as concluded in Section 3.1.

Table 3. Prediction Results on Validation Set for the Ablation Study[a]

model	SEN (%)	SPE (%)	ACC (%)	MCC (%)
No-NT	84.16	74.21	79.19	58.66
No-DB2	83.71	**81.90**	82.81	65.62
No-CNN	84.62	79.64	82.13	64.33
No-Resnet	86.43	78.28	82.35	64.92
NBCR-ac4C	**91.40**	76.02	**83.71**	**68.23**

[a]The highest score in each column is shown in bold.

On the testing set (Table 4), the Nucleotide Transformer's removal decreases MCC by 12.04% and ACC by 6.06%. For

Table 4. Prediction Results on Test Set for the Ablation Study[a]

model	SEN (%)	SPE (%)	ACC (%)	MCC (%)
No-NT	80.80	74.09	77.45	55.02
No-DB2	**89.31**	74.64	81.97	64.65
No-CNN	79.71	**84.78**	82.25	64.58
No-Resnet	84.78	79.89	82.34	64.75
NBCR-ac4C	84.96	82.07	**83.51**	**67.06**

[a]The highest score in each column is shown in bold.

No-DB2, No-CNN, and No-ResNet18, the MCC decrease is about 2.5%, and the ACC decreases by less than 2%. From the perspective of AUROC on the testing set (see in Figure 3), the

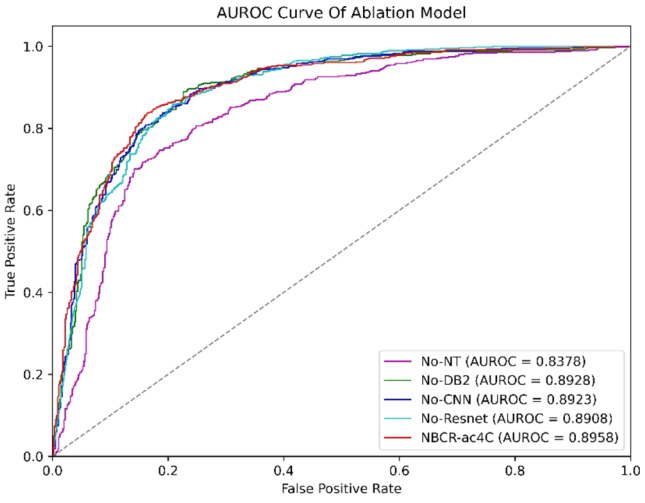


Figure 3. AUROC curves on the test set for the ablation study.

impact of the Nucleotide Transformer in feature representation on the detection performance is also higher than the other three modules. The optimal AUROC of 89.58% is achieved only when both feature representation (Nucleotide Transformer and DNABERT2) and deep learning components (CNN and ResNet18) are adopted together.

Regarding SEN on the testing set, NBCR-ac4C scores lower (84.96%) than No-DB2 (89.31%), indicating that DNABERT2 might have a limited or negative effect. This discrepancy is not observed in the validation set, possibly due to data set partitioning (Section 2.1) which may make testing set samples easier to distinguish. The SPE on the testing set also revealed

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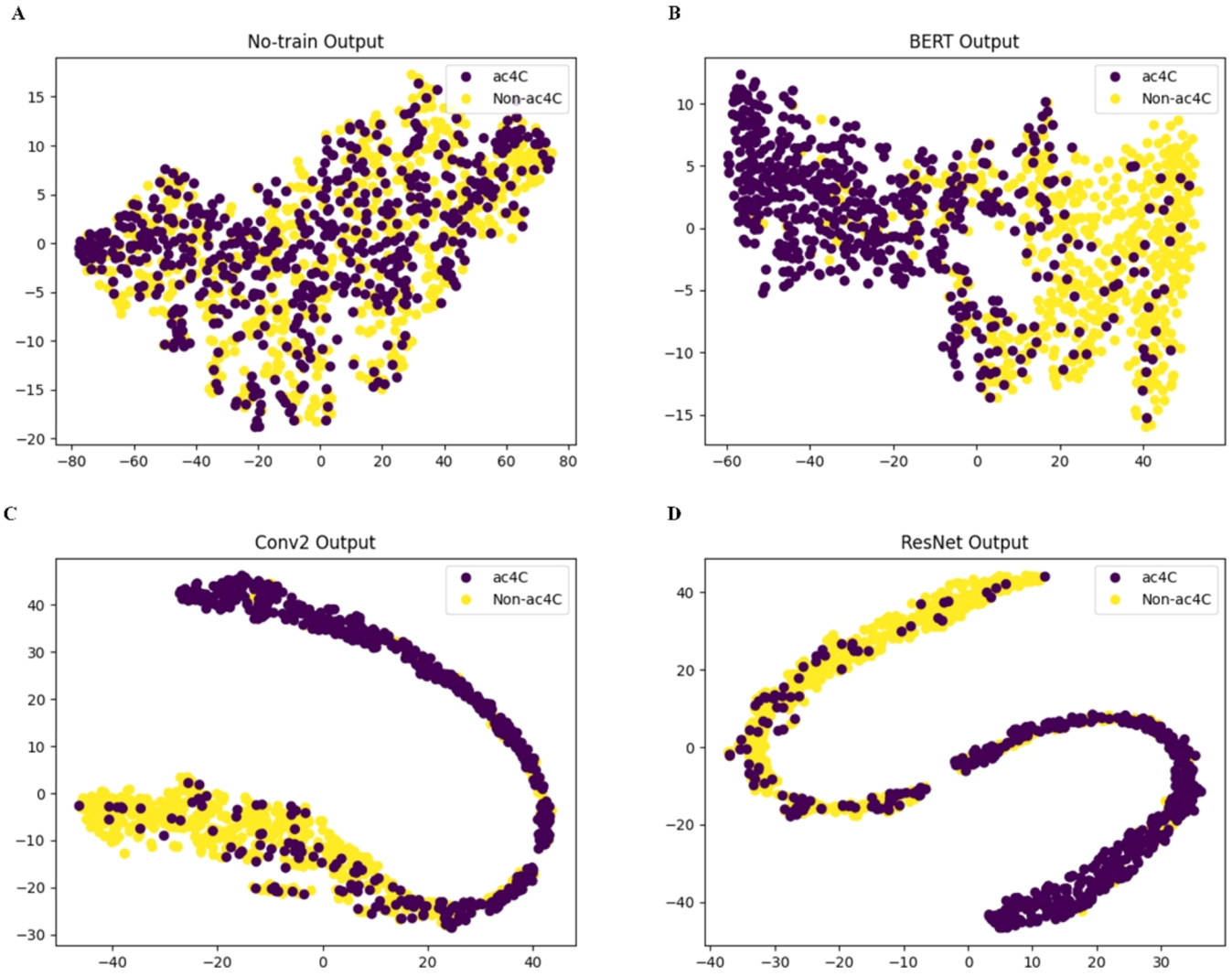


Figure 4. t-SNE results of different parts in NBCR-ac4C.

that NBCR-ac4C could not achieve optimal results. It confirms that the model designed in this study, through BERT-based feature presentation and CNN, ResNet18, etc., has learned the features and semantic knowledge of ac4C site samples sufficiently. At the same time, its ability to identify non-ac4C site samples is relatively insufficient (also related to the random partition of the data). However, in bioinformatics or medicine, SEN is more critical than SPE due to the importance of correctly identifying positive samples.

3.3. Visualization with t-SNE. This experiment utilizes t-SNE[32] for visualization to explore the roles of various components (feature representation, etc.) in the NBCR-ac4C model for distinguishing ac4C sites. Specifically, t-SNE is applied to visualize feature vectors after preprocessing with the Nucleotide Transformer and DNABERT2, after extracting shallow semantic knowledge through the CNN layer, and after obtaining deep knowledge by ResNet18.

The visualization results (Figure 4A–D) depict the initial ac4C site data set's random distribution (Figure 4A). Following feature representation, ac4C and non-ac4C sites show clustering tendencies (Figure 4B), indicating initial discriminability in the feature space generated by the BERT-based pretraining models. Results after further feature extraction via CNN and ResNet18 are similar but exhibit more pronounced discriminability between ac4C and non-ac4C sites (Figure 4C and D). Notably, ResNet18 results in slightly higher data clustering, potentially due to its ability to capture deeper semantic features of nucleotide sequences. Overall, as the NBCR-ac4C model's network complexity increases, t-SNE plot discriminability also improves, visually underscoring the importance of different parts in identifying ac4C sites.

3.4. Comparison with State-of-the-Art Models. Based on the results of the ablation study and t-SNE visualization, the effectiveness of the NBCR-ac4C model is validated. To further evaluate its performance, this experiment compares it with well-known ac4C detection models, including PACES (with three variants based on PSDSP features, KNF features, and a weighted model combining both),[33] XG-ac4C,[34] iRNA-ac4C,[9] ac4C-AFL, and LSA-ac4C. Of these, LSA-ac4C is considered the SOTA. The iRNA-ac4C combines minimum redundancy maximum relevance and incremental feature selection strategies to manually define the optimal feature subset. The LSA-ac4C focuses primarily on model construction, which establishes a deep learning model with a dual-layer long short-term memory network and a self-attention mechanism to predict ac4C sites.

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Table 5 presents NBCR-ac4C's performance compared to all baselines on the independent test set. The proposed model achieves the highest ACC, MCC, and AUROC values of 83.51, 67.06, and 89.58%, respectively, indicating a superior ac4C site identification ability. Moreover, the AUROC results demonstrate the model's ability to maintain a high true positive rate and a low false positive rate across different decision boundaries. XG-ac4C achieves the highest SEN of 92.57% but falls short in SPE at less than 60%, far below NBCR-ac4C's 82.07%. Additionally, iRNA-ac4C exhibits the highest SPE (82.91%), slightly higher than that of NBCR-ac4C, but with an 8.26% lower SEN. Overall, the proposed model demonstrates the best overall performance in determining ac4C sites.

Table 5. Prediction Results on the Test Set, Compared with Different Methods[a]

model	SEN (%)	SPE (%)	ACC (%)	MCC (%)	AUROC (%)
PACES-PSDSP	74.82	75.91	75.36	50.73	82.87
PACES-KNF	82.60	76.63	79.62	59.35	86.92
PACES	79.71	77.90	78.80	57.62	86.48
XG-ac4C	**92.57**	59.78	76.18	55.42	87.13
iRNA-ac4C	76.70	**82.91**	79.81	59.70	88.00
ac4C-AFL	84.40	80.30	82.30	64.70	89.50
LSA-ac4C	87.13	78.26	82.70	65.66	89.53
NBCR-ac4C	84.96	82.07	**83.51**	**67.06**	**89.58**

[a]The highest score in each column is shown in bold.

We believe that this performance advantage stems from the BERT-based feature representation and the integration of CNN and ResNet18 in the deep neural network. The former leverages pretrained models such as Nucleotide Transformer and DNABERT2 precisely capture contextual information among sample features, to address the limitations of manual feature extraction methods in iRNA-ac4C.[9] The latter (CNN and ResNet18) extracts shallow and deep semantic knowledge efficiently from preprocessed feature vectors of nucleotide sequences. The residual network alleviates gradient vanishing during training and facilitates cross-layer feature fusion, which enables efficient learning across different levels of nucleotide sequences compared to LSTM and self-attention (also proved by the conclusion of Section 3.1).

4. CONCLUSIONS

In this study, we develop a deep learning method, NBCR-ac4C, using BERT-based pretrained models to predict ac4C sites in nucleotide sequences accurately. The Nucleotide Transformer and DNABERT2 capture complex sequence dependencies and context. A deep neural network with CNN and ResNet18 then extracts shallow and deep semantic knowledge, transforming input samples into 1D vectors that encode positional and feature information. These vectors are fed through a fully connected layer to predict ac4C or non-ac4C sites. Extensive experiments show that NBCR-ac4C outperforms the current SOTA model, LSA-ac4C, in ACC, MCC, and AUROC, achieving values of 83.51, 67.06, and 89.58%. This indicates superior performance in identifying ac4C sites with high true positive and low false positive rates across decision boundaries. Ablation and visualization experiments reveal that each component of NBCR-ac4C positively impacts prediction, with feature representation effectively capturing context and the deep network fusing cross-layer features for precise ac4C site prediction. Based on BERT models pretrained on genomic corpora, the NBCR-ac4C model has strong generalization potential. Through transfer learning and fine-tuning, it can be adapted to predict other key RNA functional sites. Our future work aims at extending this work to other RNA modifications recognition.

■ ASSOCIATED CONTENT

Data Availability Statement

The data set and code for NBCR-ac4C are available at https://github.com/2103374200/NBCR.

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H.W.Y. and H.Y. contributed equally to this work. H.W.Y. and H.Y. performed the computational experiments. H.W.Y., H.Y., and B.Y.D. wrote and revised the manuscript. Z.Y. and G.F. conceived the study and supervised the research. All authors have given approval for the final version of the manuscript.

Funding

This work was supported by the National Natural Science Foundation of China (62302148) and the Hebei Natural Science Foundation (F2024202076).

Notes

The authors declare no competing financial interest.

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